

Supplementary Material

Are Archived NBA Consensus Moneylines Calibrated? A Pre-Specified Test of Favorite Pricing Across 15,351 Games

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Everything here is generated by the replication materials at <https://github.com/jacobburdier05/nba-moneyline-calibration> and can be regenerated with a single command. Section references are to the main paper.

S1. Shin's normalization equals the additive rule for a two-outcome book

Section 5.4 of the main paper reports that Shin (1993) and the additive rule give identical results. Because that claim is unusual and affects how Table 5 should be read, the derivation is given here rather than asserted. Every step is verified against the sample by `src/shin_equivalence_proof.py`.

Let q_1 and q_2 be the raw implied probabilities of the two sides and $P = q_1 + q_2$ the booksum, with P greater than one. Shin sets

$$pi_i = [\text{sqrt}(z^2 + 4(1 - z) q_i^2 / P) - z] / (2(1 - z))$$

with z chosen so the pair sums to one. The additive rule sets $pi_i = q_i - (P - 1)/2$.

S1.1 Invert for z

Rearrange to $2 pi (1 - z) + z = \text{sqrt}(z^2 + 4(1 - z) q^2 / P)$ and square both sides. The z -squared terms cancel, and dividing through by $4(1 - z)$, nonzero for z below one, leaves

$$pi^2 + z pi (1 - pi) = q^2 / P, \quad \text{so} \quad z = (q^2 / P - pi^2) / (pi (1 - pi))$$

S1.2 Require both sides to return the same z

Shin's z is a single property of the book, so the expression must give the same value for each side. Because the pair sums to one, $1 - pi_1$ equals pi_2 and $1 - pi_2$ equals pi_1 , so both denominators equal $pi_1 pi_2$ and the requirement reduces to equal numerators:

$$(q_1^2 - q_2^2) / P = pi_1^2 - pi_2^2, \quad \text{that is} \quad (q_1 - q_2)(q_1 + q_2)/P = (pi_1 - pi_2)(pi_1 + pi_2)$$

Since $q_1 + q_2 = P$ by definition and $pi_1 + pi_2 = 1$ by the constraint, this collapses to $q_1 - q_2 = pi_1 - pi_2$. Shin's rule preserves the difference of the raw probabilities.

S1.3 Solve

Two linear conditions now pin the pair down uniquely: the sum is one and the difference is $q_1 - q_2$. Therefore $\pi_1 = (1 + q_1 - q_2)/2$. The additive rule gives $q_1 - (q_1 + q_2 - 1)/2$, which is the same expression. The two rules coincide.

S1.4 Numerical verification

Check	Maximum absolute error
S1.2, difference preserved	3.3e-16
S1.3, closed form $\pi_1 = (1 + q_1 - q_2)/2$	2.8e-16
Equivalence, $ \text{Shin} - \text{additive} $	2.8e-16
Sum constraint, $ \pi_1 + \pi_2 - 1 $	2.2e-16
Recovered z consistent across both sides	1.5e-14

Table S1. Verified across the 15,350 games with a positive margin. All errors are at the limit of double precision.

The recovered insider share z ranges from 0.0037 to 0.0718 with a mean of 0.0384, so these are genuine interior Shin solutions rather than a degenerate corner. One game in the archive is quoted with a booksum below one, where Shin's model has no valid solution; it is excluded here and falls back to proportional in Table 5 of the main paper.

S1.5 Scope, and an implementation note

The equivalence does not mean the two rules express the same idea. Shin's is derived from a bookmaker pricing against insider money and is intended to load margin onto longshots; the additive rule is a mechanical convention with no behavioral content. They coincide only because a two-outcome market leaves a single degree of freedom once the sum constraint and the difference-preserving property are both imposed. With three or more outcomes they separate.

One implementation note, recorded because it produced wrong numbers before it was caught. Evaluated directly, the numerator $\sqrt{z^2 + a} - z$ is a difference of nearly equal quantities as z approaches one, and in double precision it loses most of its significant digits, sending the root finder to values whose probabilities sum to roughly 0.98 rather than one. Multiplying by the conjugate gives the algebraically identical and numerically stable form $\pi = 2 q^2 / [P (\sqrt{z^2 + a} + z)]$, which is what the code uses.

S2. Exact power, and the normal approximation

The minimum detectable effect quoted in the main paper is the conventional two-sided normal approximation, not an exact inversion of the Poisson-binomial power function. Because the two are not guaranteed to agree, `src/power_curve.py` computes the exact version: it builds the exact Poisson-binomial distribution by discrete Fourier transform of its characteristic function, using the conjugate symmetry so only half the frequencies are evaluated, finds the exact two-sided rejection region under the null, and inverts exact power at 80 percent.

Quantity	Value
Exact rejection region	wins $\leq 5,425$ or $\geq 5,554$
Exact size (discreteness, not .05)	.0484
MDE, normal approximation	1.3281 pp
MDE, exact Poisson-binomial	1.3242 pp
Difference	0.0039 pp
Exact null mean vs analytic sum(p)	5489.4484 vs 5489.4484
Exact null SD vs sqrt(sum p(1-p))	32.4259 vs 32.4259

Table S2. The exact null mean and standard deviation match their analytic counterparts to four decimal places, which is an independent check on the primary test's variance.

True gap (pp)	Exact power	Normal approximation
0.50	17.75%	18.39%
1.00	55.56%	55.94%
1.33 (MDE)	80.35%	80.11%
1.50	88.97%	88.57%
2.00	99.00%	98.81%
3.01 (break-even)	>99.99%	>99.99%

Table S3. Exact against approximate power. The largest disagreement anywhere in the range is 0.64 percentage points of power.

For comparison, the illustrative narrow bucket in section 1 of the main paper, 74 games at an assumed win probability of .85, has 11.2 percent power against the same 3.01-point break-even margin.

S3. The withdrawn literature range

An earlier draft of this paper stated that favorite bias in the range commonly claimed, roughly 1.7 to 5.1 percentage points, had been reported in prior work, and

Figure 3 marked both values as the smallest and largest effects in the cited literature. Neither number could be traced to any source in that citation set. Both were constants with no citation behind them. This section records what the attempt to source them found, because the finding is substantive rather than clerical.

Six of the eight cited papers report rates of return or point-spread cover rates, not win-probability calibration gaps. The two quantities are not interchangeable without attaching the odds: a return R on a favorite priced at raw probability q corresponds to a calibration gap of the vig share plus q times R . At this sample's prices a -5.5 percent return corresponds to roughly -1.6 points and a -23 percent return to roughly -16, so the classic return literature, honestly converted, spans a far wider range than 1.7 to 5.1 and points the opposite way, because it concerns parimutuel horse racing rather than a two-outcome book.

Newall and Cortis (2021), the one literature review in the citation set and the obvious place such a range would live, is entirely directional and reports no effect magnitudes. Levitt (2004) and Paul and Weinbach (2008) report point-spread cover rates against a 52.38 percent break-even, not moneyline calibration. Moskowitz (2021) explicitly excludes favorite-longshot bias from its scope. Angelini and De Angelis (2019) study three-outcome football markets. Griffith (1949) and Thaler and Ziemba (1988) concern parimutuel horse racing.

The moneyline branch rests largely on Woodland and Woodland (2001), whose conversion method was questioned by Berkowitz, Depken and Gandar (2018) on the grounds that it overstates bookmaker commission and understates underdog win probability, and whose authors reported in 2011 that the effect is sustained for the first three seasons but disappears over the following seven as the market converges to efficiency.

The marks were therefore removed rather than re-sourced, since no defensible replacement exists in those units, and the power argument in the main paper is made against this sample's own break-even threshold instead. One coincidence is worth recording: 1.7 is numerically almost identical to this paper's own largest bucket deviation, +1.73 points in the .75 to .80 bucket.

S4. Season-by-season results

Favorites above .70 vig-free implied probability, by season. No season rejects; the smallest p is .107. The Cochran Q heterogeneity statistic over these thirteen strata is 8.37 on twelve degrees of freedom, $p = .756$, with I-squared at 0 percent.

Season	n	Observed	Implied	Gap (pp)	z	p
2007-08	601	81.70%	81.00%	+0.70	0.44	.658
2008-09	565	82.48%	80.31%	+2.17	1.32	.188
2009-10	529	81.66%	80.28%	+1.38	0.81	.417
2010-11	536	82.46%	79.94%	+2.52	1.48	.140
2011-12	442	81.90%	80.00%	+1.90	1.01	.311
2012-13	525	80.76%	79.77%	+0.99	0.57	.567
2013-14	530	80.75%	80.43%	+0.33	0.19	.846
2014-15	585	80.68%	80.42%	+0.27	0.16	.869
2015-16	546	82.05%	81.51%	+0.54	0.33	.741
2016-17	490	77.96%	80.78%	-2.82	-1.61	.107
2017-18	513	78.75%	79.38%	-0.62	-0.35	.723
2018-19	539	81.08%	80.07%	+1.00	0.59	.555
2019-20	439	77.68%	79.02%	-1.34	-0.70	.484

Table S4. Exploratory, not pre-specified, and not corrected for multiplicity.

S5. Reproducing every number

Clone the repository, install the requirements, and run `bash run_all.sh`. The pipeline runs in eight steps and takes about three minutes. Data verification runs first and halts the pipeline on failure.

Step	Script	Produces
1	<code>verify_data.py</code>	full-sample cross-validation, schedule and outcome checks
2	<code>build_dataset.py</code>	exclusions, probabilities, favorite side, break-even
3	<code>primary_analysis.py</code>	primary test, calibration regression, buckets, returns
4	<code>robustness.py</code>	seasons, Cochran Q, extreme tail, season bootstrap
5	<code>figures.py</code>	Figures 1 and 2
6	<code>normalization_robustness.py</code>	five vig-removal rules (Table 5)
6	<code>dependence_robustness.py</code>	cluster-robust variance, team bootstrap (Table 4)
7	<code>power_curve.py</code>	Figure 3, Table 1, exact power (Tables S2 and S3)
8	<code>shin_equivalence_proof.py</code>	S1 verification (Table S1)

Table S5. Every statistic in the main paper is printed to the console and written to `results/` as JSON or CSV.

A separate script, `src/fetch_source_data.py`, re-downloads the complete upstream files, prints their SHA-256 hashes, and confirms that every retained column in the

local subsets matches the original row for row. The frozen analysis plan is at preregistration/analysis_plan.md, with a provenance note stating when it was transcribed into the repository and what evidence does and does not exist for the date it was frozen. A running record of every correction made to this paper, including the two documented in S2 and S3, is at docs/errata.md.